§7 Collapse, blockage and collapse path sets — part 5 of 5 (inflow tendency; connection to the organizational ACS)

Source: `paper/section7_collapse.tex`, lines 3627–4834 (`subsec:inflow_paths`: inflow to pre-collapse states, described as a tendency; `subsec:to_section8`: connection to the organizational ACS), at commit e6e12cd, with the one-line change to line 2048 recorded in the ledger. Status: draft. Part 4 is `07_collapse_4.md`; this is the last part of §7. §7 defines no assumption. The Japanese manuscript is authoritative; this is a summary, not a translation, and adds no definition, assumption or result.

Inflow to pre-collapse states

`Reference: subsec:inflow_paths`

The subsection organizes, at a descriptive level, how changes in time-indexed quantities of the individual ACS Σ_i and changes in its relations to external objects *may be related to* the descriptive pre-collapse state bands $\mathsf{Z}_1, \dots, \mathsf{Z}_4$. This is called an *inflow tendency*. The subsection keeps three things apart: changes of time-indexed quantities or relation states, the descriptive pre-collapse state bands, and membership in the blockage and collapse path sets. An inflow tendency does not denote a deterministic transition from a quantity or a relation state to some Z_j, to blockage, or to collapse.

Status of the inflow tendency

Not a formal transition relation (`rem:inflow_is_descriptive`). The words “inflow” and “inflow tendency” introduce no new transition relation, vector field, distance, gradient flow or probabilistic transition on the state space X. No formal transition rule from a state described as Z_j to one described as Z_k is defined. Since Phase 1 gives the trajectory record space $\mathfrak{P}$ no distance or topology in general, no formal quantity such as “distance to $\mathfrak{P}_{\mathrm{blk}}$” or “approach to $\mathfrak{P}_{\mathrm{col}}$” is introduced either.

A description of classification C (`rem:inflow_description_level`). The time-indexed quantities of §5, the relation descriptions of §6 and the bands Z are objects on the theory’s side of description. The subsection does not require Σ_i itself to hold or compute an internal variable or label saying “I am flowing into a pre-collapse state”. Nor is it a Phase 1 requirement that M2 or another mechanism recognize the inflow tendency, Z_j, or membership in $\mathfrak{P}_{\mathrm{blk}}$ or $\mathfrak{P}_{\mathrm{col}}$ as a truth value.

Connection with time-indexed quantities

Reference to the quantities of §5 (`rem:internal_inflow_types`). The organizing tuple $\mathrm{Desc}_t = (\boldsymbol{\beta}_t, d_{\mathrm{eff},t}, \pi_{\tau,t}, \hat{\tau}_t, h_{\mathrm{psl},t}, h_{\mathrm{mem},t}, h_{\mathrm{sens},t}, h_{\mathrm{eff},t})$ is referred to; it is a classification-C notation for organizing, not an independent state vector. When these quantities change over a finite time interval in a direction described as enlarging the difference from normal operation and the reference band Z_0, a relation to Z_1–Z_4 may be considered. This paper does not state, as a general consequence, the assignment to a particular Z_j from a particular value, sign, threshold or finite-window difference. The relation of health quantities to blockage, collapse and the bands follows `rem:s5_health_noncollapse` of §5 and `rem:health_not_zone_identity` of §7.

$d_{\mathrm{eff},t}$ and $\mu_{\mathrm{eff},t}$ (`rem:d_eff_not_mu_eff`). The effective development freedom $d_{\mathrm{eff},t}$ of §5 and the response diversity $\mu_{\mathrm{eff},t}$ of §3 (classification C; the number of feasibility-profile equivalence classes of feasible actions over $E_{\mathrm{req}}(x_t)$ at (x_t, e_t), a local one-step quantity) are different formal objects. The following are not assumed: $d_{\mathrm{eff},t} = \mu_{\mathrm{eff},t}$; $d_{\mathrm{eff},t} \downarrow \Rightarrow \mu_{\mathrm{eff},t} \downarrow$; $d_{\mathrm{eff},t} = 0 \Rightarrow \mu_{\mathrm{eff},t} < 2$, each without additional conditions. The defining condition of the local FR does not use $\mu_{\mathrm{eff},t}$: local FR is judged on $\mathcal{D}_{\mathrm{FR}}$ by non-emptiness of the feasible action set. This paper therefore does not state that local FR is false from $\mu_{\mathrm{eff},t} < 2$ alone, and a decrease of $d_{\mathrm{eff},t}$ is not itself treated as failure of the local FR condition.

Time-horizon quantities (`rem:time_horizon_not_collapse`). A decrease of $\pi_{\tau,t}$ or $\hat{\tau}_t$ may, in some systems, be related to a shrinking margin of prediction, evaluation and response. This paper does not adopt $\hat{\tau}_t = 0 \Rightarrow A_{i,t} = \emptyset$ or $\hat{\tau}_t = 0 \Rightarrow \mathrm{Col}_{i,t} = 1$ in general, and does not state the assignment to a particular Z_j from a particular value of $\pi_{\tau,t}$ as a general consequence.

Bias strength (`rem:beta_not_fixed_collapse_direction`). For the bias strength $\beta_{\iota,t}$ ($\iota \in \mathcal{J}_\beta$), neither the attribution division nor a direction is read from its value. The subsection does not fix either an increase or a decrease of $\beta_{\iota,t}$ universally as a "collapse direction": the relation to pre-collapse states depends on the reference standard and scale of the bias ι, the readout rule and carrier transitions, and the combination with other quantities. Phase 1 therefore places no general implication from $\beta_{\iota,t} \uparrow$ to $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or to $\sigma \in \mathfrak{P}_{\mathrm{col}}$.

Connection with the time-indexed satisfaction predicates of the three principles

Internal quantities are not the predicates (`rem:internal_quantities_not_principles`). Changes in the quantities of §5 may coexist or be related, in some systems, with state changes at which a time-indexed satisfaction predicate of SM, EU or FR is false. This paper does not take $h_{q,t}$, $d_{\mathrm{eff},t}$, $\boldsymbol{\beta}_t$ or $\hat{\tau}_t$ to be the truth value of any such predicate. The SM predicate at time t is judged by $\Pi(x_t) \geq 0$. The EU predicate is defined in the domain of §3 and judged by satisfaction of the required environmental factors in $\mathrm{Crit}^{(1)}(x_t)$. The local FR predicate is defined on $\mathcal{D}_{\mathrm{FR}}$ and is equivalent there to $A_{i,t} \neq \emptyset$. $\mu_{\mathrm{eff},t}$ is a classification-C quantity and not the truth value of local FR. A change in the quantities of §5 is therefore not used to replace the judgment of the predicates.

Falsity of a predicate and path-set membership (`rem:principle_failure_not_path_membership`). This paper does not state $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$ as a general consequence from the SM predicate being false alone (`rem:sm_wf_not_blk_path`, `rem:principles_not_col_path`). The local FR predicate, on the other hand, is equivalent to $A_{i,t} \neq \emptyset$ on $\mathcal{D}_{\mathrm{FR}}$ (`rem:internal_quantities_not_principles`). If it is false at t, then $A_{i,t} = \emptyset$, and when $t \in I_{\mathrm{ACS}}(\sigma)$, `def:blk_path_set` gives $\sigma \in \mathfrak{P}_{\mathrm{blk}}$. The relation between the EU predicate being false and blockage is treated in §3, and this remark does not presuppose it. This paper does not state $\sigma \in \mathfrak{P}_{\mathrm{col}}$ as a general consequence from any of the three predicates being false alone. Even when the subsection speaks of a descriptive connection between internal quantities and a false predicate, it is not treated as a deterministic path to blockage or collapse.

Connection with changes originating in relations

Relation descriptors (`rem:relational_inflow`). A change over time in a relation state with external objects may, in some systems, be related to the internal state x_t, the quantities of §5, satisfaction of required environmental factors, or the feasible action set. The subsection allows such relation-originated change to form a relation with the descriptive pre-collapse bands. This paper does not state, as a general consequence, a particular Z_j, blockage or collapse from a particular relation category, relation-descriptor component or relation direction.

Relation categories are not labels of causes of collapse (`rem:relation_category_not_collapse_cause`). Even where §6 introduces relation categories, the subsection does not fix a category as one that "causes collapse". Relations described by the same category name can affect an ACS differently, depending on the concrete content, duration, direction, object state, environment state, internal state and combination with other relations. No general implication from a relation category to $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$ is adopted.

Negative relation quality (`rem:negative_Q_not_collapse`). $Q_{i \leftarrow j,t} < 0$ is a negative-side evaluation on the quality index. This paper does not state, from it alone, a decrease direction of $\Pi(x_t)$, blockage, loss of well-formed realization, or collapse. Hence it does not state $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$ from $Q < 0$. A further model that connects the accumulation of relation quality over time, or integrated evaluation of several relations, to collapse paths is not introduced here.

Environment coupling and realized relation change (`rem:env_coupling_not_relational_inflow`). The structural environment coupling condition of §3 is a background condition on the structural possibility of coupling between an ACS and the external environment. It is positioned as a premise and background structure of EU, but it is not the EU predicate itself. The relation-originated change treated here describes the realized relation state with external objects and its change over time. The structural coupling condition, the realized relation state and the relation-originated inflow tendency are therefore distinct objects and are not identified. A case in which the coupling condition holds and an unfavorable relation state is realized is not excluded, and failure of the condition does not immediately yield blockage or collapse.

Inflow tendency and the state bands

No unique band (`rem:inflow_not_unique_zone`). Even when a change in internal time-indexed quantities or a relation state is observed, Phase 1 defines no universal map that determines one of $\mathsf{Z}_1, \dots, \mathsf{Z}_4$ from it. The same change may relate to a mild-impairment band in one system, remain within normal operation in another, and relate to several bands at once in a third; none of these is excluded. This agrees with §7 not having defined the bands as a partition of the state space or as a total order.

No monotone departure (`rem:inflow_not_monotone`). The word "inflow tendency" does not mean a monotone sequence passing through the bands in index order from Z_0 to Z_5. Phase 1 does not exclude a return to normal operation or the reference-band side after a change, a move to a different descriptive band, or reaching the collapse terminal without passing through an explicit pre-collapse band.

Inflow tendency and path-set membership

Distinct conditions (`rem:inflow_vs_membership`). The inflow tendency and membership of a trajectory record in $\mathfrak{P}_{\mathrm{blk}}$ or $\mathfrak{P}_{\mathrm{col}}$ are different conditions. Even if a decrease of a health quantity, a change of $d_{\mathrm{eff},t}$, of a time-horizon quantity, of bias strength, or of a relation state shows a relation to pre-collapse states in some record, this paper does not state blockage or collapse within finite time as a general consequence. Hence it does not state $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$ from an inflow tendency.

No reverse inference (`rem:membership_not_unique_inflow`). Conversely, this paper does not state, from $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$, that the trajectory had earlier passed through a particular health decrease, a particular change of $d_{\mathrm{eff},t}$, a particular relation category or a particular change of Q. The concrete mechanism leading to blockage or collapse can depend on the system's internal dynamics, the environment realization (the sequence of background components and g_O), the relation state, external actions and other conditions. Membership in a path set does not mean a unique identification of a causal mechanism.

Collapse without a pre-collapse band (`rem:inflow_and_direct_collapse`). As fixed in `subsec:collapse_def` and `subsec:col_path_set`, $\mathrm{Col}_{i,t} = 1$ does not require prior passage through Z_1–Z_4. The "inflow to pre-collapse states" of this subsection therefore does not cover all paths that can lead to collapse. Collapse-reaching paths that do not explicitly pass through a pre-collapse band lie outside this subsection's organization, but they are not excluded from the Phase 1 trajectory record space.

Connection with M2 and TIPA

Not a direct input of M2 (`rem:path_sets_m2_connection`). $\mathfrak{P}_{\mathrm{blk}}$, $\mathfrak{P}_{\mathrm{col}}$, $\mathsf{Z}_1, \dots, \mathsf{Z}_4$ and the inflow tendency are objects on the theory's side of description. It is not required that M2 hold or compute these sets, membership in them, band labels, or a "degree of inflow" as internal input. The formal input type, evaluation map and Priority of M2 are the responsibility of §4 and are not redefined here. An implementation in which feasibility-related state changes on a trajectory in a path set are reflected in information available to M2 (sensing, integration, internal model) is allowed. This paper does not state a particular M2 output or Priority value as a general consequence from $\sigma \in \mathfrak{P}_{\mathrm{blk}}$ or $\sigma \in \mathfrak{P}_{\mathrm{col}}$.

Phase 1 limits and summary

Limits (`rem:inflow_phase1_scope`). Phase 1 only organizes, descriptively, that changes in internal time-indexed quantities and in relation states with external objects may be related to pre-collapse states, for the trajectory records of a single individual ACS Σ_i. In general it does not introduce: a distance or potential to pre-collapse states; an inflow speed or rate; a probability of blockage or collapse; a universal threshold map from a quantity to a Z band; a causal law from a relation category to collapse; a sufficient condition for path-set membership drawn from an inflow tendency. Introducing these requires additional system-dependent assumptions, topological, measure or probabilistic structure, or empirical grounds. Their precise formalization is left to Phase 2 and later (L9 of `subsec:limitations_list`).

Summary (`rem:inflow_summary`). The relations kept are *change of time-indexed quantities* ⇢ *possible relation to pre-collapse states*, and *change of relation state* ⇢ *possible relation to pre-collapse states*, where ⇢ denotes a descriptive connection and not a formal implication. This paper does not state a classification into a particular Z_j from a change alone, nor membership of σ in $\mathfrak{P}_{\mathrm{blk}}$ or $\mathfrak{P}_{\mathrm{col}}$ from an inflow tendency alone.

Connection to the organizational ACS

`Reference: subsec:to_section8`

The subsection connects the individual-level concepts of §7 to the discussion of §8, where the organization itself has ACS structure. The individual-level concepts of §7 are not lifted to the organizational level as they are.

Group of ACSs versus organizational ACS

A group is not by itself an organizational ACS (`rem:acs_group_vs_org_acs`). Several individual ACSs $\Sigma_1, \dots, \Sigma_n$ together with interactions or a relational structure among them may form the substrate of a group of ACSs in the sense of §8. This paper does not state, from the existence of several individuals or of a relation network alone, an organization-level structural self, organization-level feasibility, or the holding of an organizational ACS. Thus *group of ACSs ≠ organizational ACS*. That an organizational ACS holds requires the organization-level ACS structure defined separately in §8; the subsection does not anticipate or redefine those conditions.

Objects carry the individual index i (`rem:individual_level_failure_objects`). $A_{i,t}$, $\mathrm{WF}_{i,t}$, $\mathrm{Col}_{i,t}$, $\Upsilon_{\mathrm{adm},i}$, $\mathrm{Recover}_{i,t}$ and the trajectory-record objects $\mathfrak{P}$, $\mathfrak{P}_{\mathrm{blk}}$, $\mathfrak{P}_{\mathrm{col}}$ of an individual Σ_i are, within the scope of §7, objects for a single individual ACS. The individual index i is therefore not dropped in order to read them as organization-level blockage, collapse or recovery. In a system with several individuals, blockage or collapse of each individual is judged first for each Σ_i separately.

Individual collapse versus organization-level states

Not identified (`rem:individual_vs_org_collapse`). Even when $\mathrm{Col}_{i,t} = 1$ holds for one member, this paper does not state, from it alone, collapse of the organizational ACS as a whole. Conversely, where an organization-level notion of collapse is treated in §8 or a later Phase, collapse at the organization level need not yield $\mathrm{Col}_{i,t} = 1$ for every member. Thus *collapse of an individual ACS ≠ collapse at the organization level*. Formalizing organization-level collapse requires conditions on the organization's own structural self, organization-level feasibility and organization-level coupling structure, and the collapse conditions of §7 would then be checked as satisfied at the organization level, under the organization-level ACS structure of §8.

Individual blockage and organization-level feasibility (`rem:individual_blockage_org_viability`). Even when $A_{i,t} = \emptyset$ holds for an individual Σ_i, this paper does not state, from it alone, blockage of another individual Σ_j, functional stoppage of the whole group, or loss of feasibility of an organizational ACS. Likewise it does not define collapse of an organizational ACS from simultaneous blockage of several individuals alone. Connecting individual-level fail-

ure states to organization-level states requires separately using the organization-level ACS structure introduced in §8.

Spread through relations

Only as a possibility (`rem:multi_agent_collapse_deferral`). When an external object is another ACS Σ_j, the Phase 1 general theory does not exclude that the actions or state changes of Σ_j take part, through a relation, in a state change of Σ_i. This paper does not state a formal law that "collapse propagates between individuals" as a general consequence. In particular it places no inter-individual propagation law such as $\mathrm{Col}_{j,t} = 1 \Rightarrow \mathrm{Col}_{i,t+k} = 1$ without additional assumptions. Treating multi-agent spread as formal dynamics needs an additional formalism describing the simultaneous time evolution of the states and relation structure of several individuals, left mainly to Phase 2 and later after the structural connection in §8 (`prob:social_dynamics`).

Relation categories and multi-agent collapse (`rem:relation_category_collapse_scope`). The relation categories and descriptors of §6 describe relations between individuals. This paper does not state multi-agent blockage, collapse propagation, or collapse of an organizational ACS as a general consequence from the existence of a particular relation category, from a negative relation quality Q, or from a large change of a relation state alone. §7 therefore does not construct in advance a causal law at the organization level or between several agents.

External recoverability in a multi-agent context

Actions by another ACS (`rem:multi_agent_recovery_scope`). An action from an ACS Σ_j to Σ_i may take part in the recovery of Σ_i. An arbitrary action by Σ_j is not taken to be $\upsilon \in \Upsilon_{\mathrm{adm},i}$: whether an external action is treated as an element of $\Upsilon_{\mathrm{adm},i}$ depends on the admissible action domain on the side of Σ_i fixed in `subsec:external_recovery`. Recovery actions performed jointly by several individuals, recovery by the organizational structure itself, and top-down actions of an organizational ACS on its members are not formalized in general here. Treating them requires additional definitions consistent with the organizational structure of §8.

Path sets in a multi-agent setting

Not introduced here (`rem:path_set_multi_agent`). $\mathfrak{P}_{\mathrm{blk}}$ and $\mathfrak{P}_{\mathrm{col}}$ are defined as subsets of the trajectory record space $\mathfrak{P}$ of a single individual ACS. Introducing a product trajectory space such as $\prod_{i=1}^{n} \mathfrak{P}_i$, or a coupled trajectory space including the time evolution of the relation network, is possible, but Phase 1 does not formally introduce such a multi-agent trajectory space. Multi-agent blockage path sets, collapse-reaching path sets, simultaneous reaching times, propagation speeds and chain-collapse sets are therefore not defined in §7. They are tasks of later extension that presuppose the organizational structure of §8.

Hand-over interface to §8

What §7 hands over (`rem:sec7_to_sec8_interface`). When §8 refers to individual-level failure states, the settled items handed over from §7 are at least the following:

1. individual blockage: $A_{i,t} = \emptyset$;
2. individual collapse: $\mathrm{Col}_{i,t} = 1$ records that (SC-1) loss of the organizational relations that establish structural identity and (SC-2) relative non-recoverability as the same individual held at some $t' \le t$; the indicator has the absorption property (C-2) within the present framework;
3. at the individual level, this paper does not state $\mathrm{Col}_{i,t} = 1$ from blockage $A_{i,t} = \emptyset$ as a general consequence;
4. individual collapse is not the same as organization-level collapse;
5. no general inclusion relation is placed between an individual's $\mathfrak{P}_{\mathrm{blk}}$ and $\mathfrak{P}_{\mathrm{col}}$;
6. changes of internal quantities and relation states are not a general sufficient condition for blockage or collapse.

§8 refers to these individual-level definitions without changing them, and defines organization-level structure separately as needed.

Boundary of responsibility

Boundary of §7 (`rem:sec7_scope_boundary`). The responsibility of §7 extends, for a single individual ACS in Phase 1, to distinguishing blockage from collapse, organizing recoverability and terminality, and describing them as sets of trajectory records. §7 does not introduce, as a general theory: conditions for an organizational ACS itself to hold; an organization-level structural self; an organization-level collapse indicator; an inter-individual collapse propagation law; a multi-agent trajectory space; multi-agent blockage and collapse path sets; an organization-level recovery operator. Of these, the holding conditions of an organizational ACS and organization-level feasibility, blockage and collapse are the responsibility of §8, while the propagation law, the multi-agent trajectory space and the multi-agent path sets are responsibilities of Phase 2 and later (`prob:social_dynamics`, `subsec:phase_outlook`). The connection from §7 to §8 is therefore limited to *handing over the settled individual-level definitions*, and the boundary that *individual definitions are not extended to the organization level without further definition* is maintained.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.